reached when the thickness of the backing is about one-fifth of the extrapolated range of the β-particles in that material.

7.5. ABSORPTION OF γ-RADIATION

The absence of charge and rest mass for γ-rays results in little interaction with the absorbing atoms and in long ranges. The number of ion pairs produced in a given path length by γ-rays is only 1—10% of that produced by β-particles of the same energy (Fig. 7.7); e.g. a 1 MeV γ-ray produces only about one ion pair per centimeter of air. As a consequence of this low specific ionization of γ-rays, the ionization is almost completely secondary in nature resulting from the action of a few high energy primary ion pairs.

7.5.1. Attenuation coefficient

Unlike heavy particles and electrons which lose their energy as a result of many collisions, γ-rays are completely stopped in one or, at most, a few interactions. For thin absorbers the attenuation of γ-rays follows Eqn (7.8), where φ is the number of photons $m^{-2}\ s^{-1}$. The proportionality factor μ is called the (*total*) *attenuation coefficient*. When it has the dimension of m^{-1} and the thickness x is expressed in meters, μ is referred to as the *linear* attenuation coefficient. The attenuation coefficient can be expressed in other ways:

$$\mu_m = \mu/\rho = \sigma_a \mathbf{N}_A/M = \sigma_e Z\, \mathbf{N}_A/M \tag{7.20}$$

where ρ is the density, M the average atomic weight, and Z the average atomic number of the absorber. $\mathbf{N}_A\rho/M$ can be replaced by N_v, by which we can define a *macroscopic absorption cross-section* Σ (cf. §11.1). Σ^{-1} is the mean free path or *relaxation length* of the radiation in the absorbing material. μ_m (in $cm^2\ g^{-1}$ when x is in centimeters) is the *mass attenuation coefficient*; σ_a is the probability of reaction between a γ-ray and the electron cloud of the absorber atom (*atomic reaction cross-section*, m^2 atom^{-1}); σ_e is the probability of the reaction of a γ-ray with a single electron of the absorber (*electron reaction cross-section*, m^2 electron^{-1}). σ_a and σ_e are analogous to the nuclear reaction cross-sections discussed later (cf. Ch. 11). In Table 7.1 only the equivalent nuclear and atomic reaction cross-sections are given.

Since a γ-ray may be removed from the beam in the first few Ångströms of its entrance into the absorber or may travel several centimeters with no interaction at all and then be removed, it is not possible to apply the range concept to γ-rays in the way that it is applied to charged particles. However, it is experimentally easy to measure the thickness of absorber necessary to remove half of the initial γ-rays (half thickness value) from a beam, or reduce it to 1/10, 1/100, etc. Figure 7.15 shows the required thickness of concrete and lead necessary to reduce γ-rays of different energies by factors of 10 (shielding thicknesses are further discussed in §7.7). The half-thickness value is

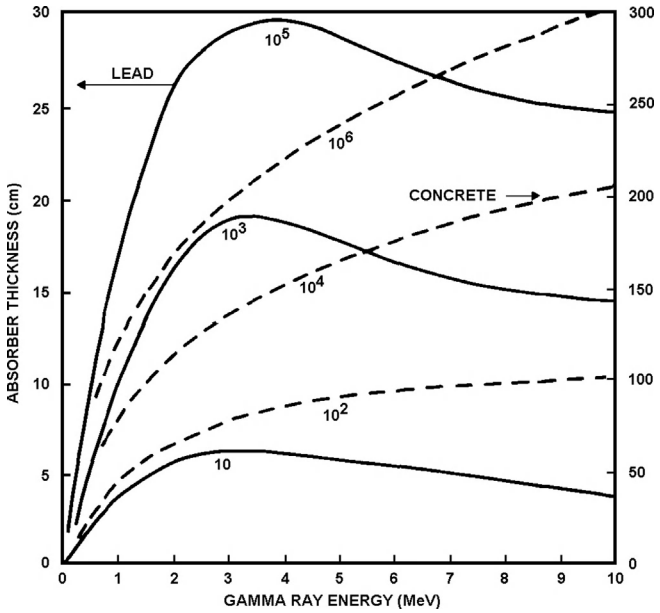


Figure 7.15 Thickness of lead and concrete reducing γ-ray flux by different powers of 10, as function of γ-ray energy. The curves are for thick absorbers and include build-up factors.

$$x_{½} = \ln(2)/\mu \tag{7.21}$$

and the 1/10 value

$$x_{1/10} = \ln(10)/\mu \tag{7.22}$$

7.5.2. Partial absorption processes

Gamma-ray absorption occurs as illustrated in Figure 7.16 by four different processes: coherent scattering, photoelectric effect, Compton effect, and pair production. For each of these processes, a partial coefficient can be expressed:

$$\mu = \mu_{\text{coh}} + \mu_{\text{phot}} + \mu_{\text{Comp}} + \mu_{\text{pair}} \tag{7.23}$$

Comparing with Eqn (7.10), μ_{phot} and μ_{pair} are absorption processes, while μ_{coh} is only a scattering process; μ_{Comp} contributes to both the absorption and the scattering terms. In Figure 7.17 the total attenuation, absorption, and the partial coefficients are given for water, aluminum, and lead as a function of the γ-ray energy. The corresponding linear coefficients are obtained by multiplying with ρ (for aluminum 2.7 g/cm^3, for lead 11.3 g/cm^3). It should be noted that the aluminum curves also can be used for absorption in concrete.

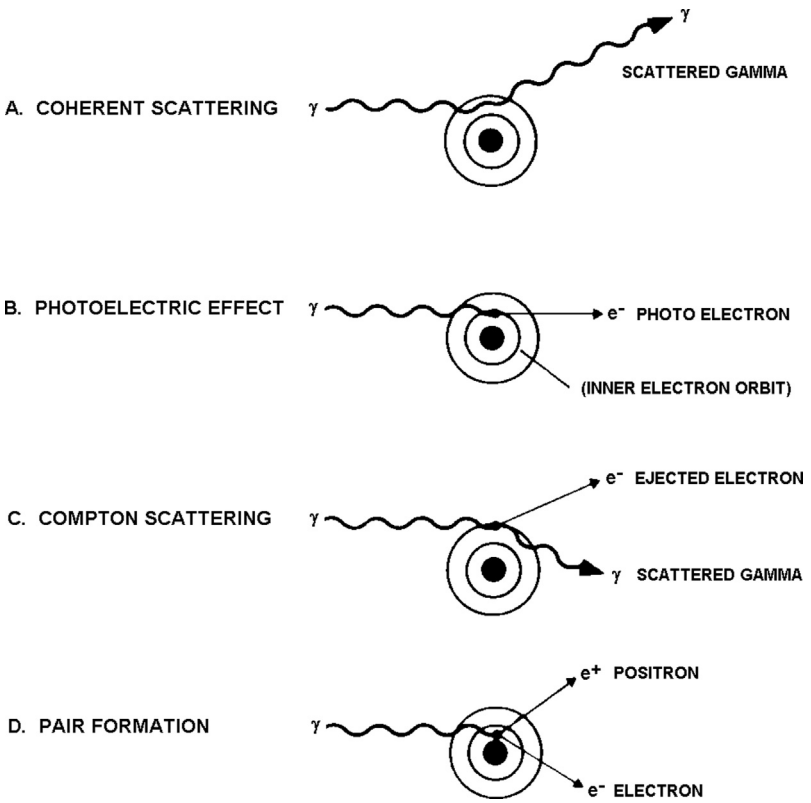


Figure 7.16 Schematic description of the four main processes accounting for γ-ray interaction and absorption.

In *coherent scattering* (also called Bragg or Rayleigh scattering, denoted σ_r in Fig. 7.17) the γ-ray is absorbed and immediately re-emitted from the atomic nucleus with unchanged energy but in a different direction. Coherently scattered radiation can give interference patterns, so the process is used for structural analysis of absorbing material in the same way as X-rays are. The probability for coherent scattering increases with the square of atomic number of the absorber and decreases with γ-ray energy. In lead, coherent scattering amounts to about 20% of the total attenuation for γ-energies of 0.1 MeV but decreases in importance for higher energy γ-rays.

In absorption of γ-rays by the *photoelectric effect* (denoted τ in Fig. 7.17 and Fig. 7.19) the photons are absorbed completely by the atom electrons. This absorption results in excitation of the atom above the binding energy of some of its orbital electrons with the result that an electron is ejected and an ion pair formed. The energy of the emitted photoelectron, E_e, is the difference between the energy of the γ-ray, E_γ, and the binding energy, E_{be}, for that electron in the atom.

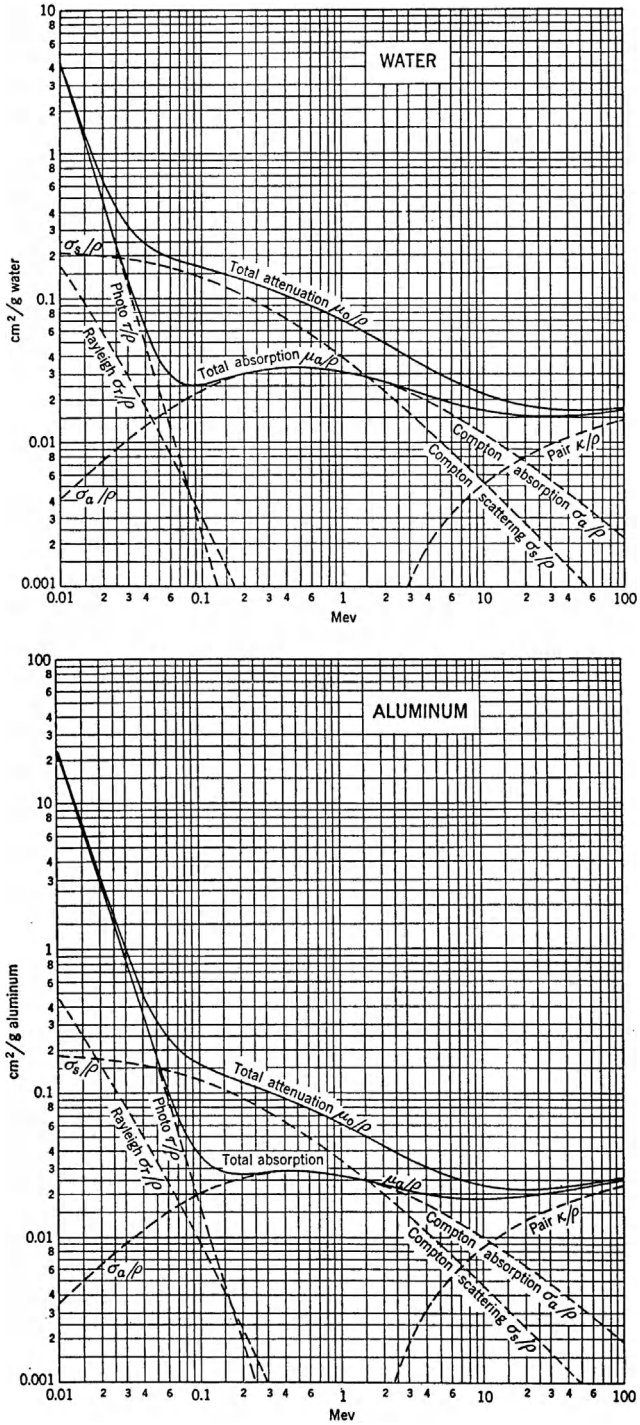


Figure 7.17 Total and partial mass absorption and attenuation coefficients for γ-rays in water, aluminum and lead. *(From R. D. Evans.)*

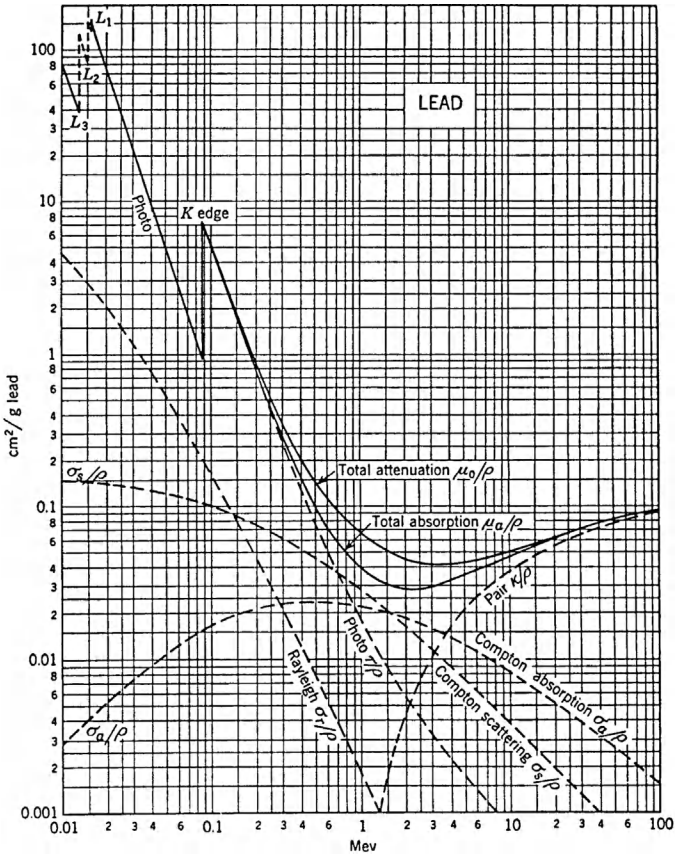


Figure 7.17 *(Continued)*

$$E_e = E_\gamma - E_{be} \tag{7.24}$$

If the photoelectron originates from an inner electronic orbital, an electron from a higher orbital moves to fill the vacancy. The difference in binding energy of the higher and the lower energy orbital causes emission of X-rays and of low energy *Auger electrons*. The process of electron cascade, accompanied by X-ray and Auger electron emission, continues until the atom is reduced to its ground state energy. The photoelectron as well as the Auger electrons and the X-rays cause extensive secondary ionization by interacting with the surrounding absorber atoms.

The probability for the photoelectric effect decreases with increasing γ-ray energy. It is largest for the most tightly bound electrons and thus the absorption coefficient for the photoelectric effect increases in the order of electron shells K > L > M >, etc. In Figure 7.17 we see that in lead it is the dominating mode of absorption up to about 0.7 MeV. Discontinuities observed in the graph of μ vs E_γ are related to the differences in

binding energies of the electrons in the different shells as the increasing γ-ray energy allows more tightly bound electrons to be emitted. These discontinuities coincide with the K, L, etc., edges observed in X-ray absorption.

Gamma-rays of higher energy, rather than interacting with the field of the whole atom as in the photoelectric effect, interact with the field of one electron directly. This mode of interaction is called the *Compton effect* after its discoverer, A. H. Compton. In the Compton effect an electron is ejected from an atom while the γ-ray is deflected with a lower energy. The energy of the scattered γ-ray, E'_γ, is expressed by the equation

$$E'_\gamma = E_\gamma - E_e \tag{7.25}$$

where E_e is the kinetic energy of the Compton electron. The probability for Compton scattering increases with target Z and decreases with E_γ. Since the Compton interaction occurs only with the most weakly bound electrons and high energy γ-rays, the binding energy of the electron is negligible compared to E_γ. The Compton electrons and scattered γ-rays have angles and energies which can be calculated from the relationships between the conservation of energy and momentum, correcting for the relativistic mass of the electrons at these kinetic energies. The scattered γ-ray may still have sufficient energy to interact further by the Compton effect, the photoelectric effect or pair production. Again, emission of X-rays and Auger electrons usually accompanies Compton interaction and extensive secondary ionization follows. Since the Compton electron can have a spread of energies, the scattered γ-rays exhibit a broad spectrum. The Compton electrons, as in the case of photoelectrons, are eventually stopped by the processes described for β-particles.

Figure 7.17 shows the division of energy between the scattered Compton γ and the Compton electron as a function of γ-ray energy. Only the energy of the electron is deposited in the absorber as the scattered γ-ray has a high probability of escape. Thus Compton electrons contribute to the (*energy*) *absorption coefficient* μ_a while the Compton γ contributes to the *total attenuation coefficient* μ through the scattering coefficient μ_s in Eqn (7.10).

The fourth mode of interaction for γ-rays with an absorber involves conversion in the Coulomb field of the nucleus of a γ-ray into an electron and a positron (Fig. 7.18). This process is termed *pair production* since a pair of electrons, one positive and one negative, is produced. The process can be considered as the inverse phenomenon of positron annihilation. Since the rest mass of an electron corresponds to 0.51 MeV, the γ-ray must have a minimum value of 1.02 MeV to interact by pair production. As the energy of the γ-ray is increased beyond this value, the probability of pair production increases (see Figs. 7.17 and 7.19, where μ_{pair} is denoted κ). The excess energy (above the 1.02 MeV) appears as the kinetic energy of the electron pair,

$$E_\gamma = 1.02 + E_{e^-} + E_{e^+} \tag{7.26}$$

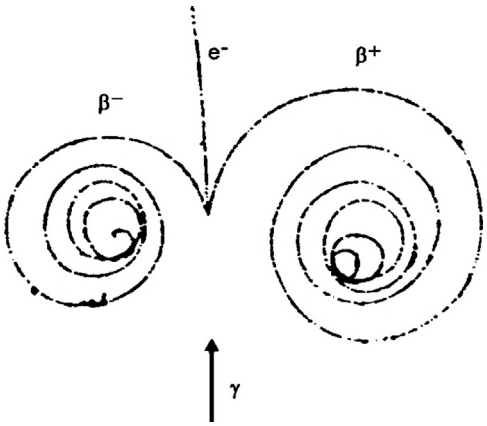


Figure 7.18 Tracks of electron pair in a H_2 bubble chamber in a strong magnetic field perpendicular to the plane of the tracks. *(Courtesy Lawrence Radiation Laboratory.)*

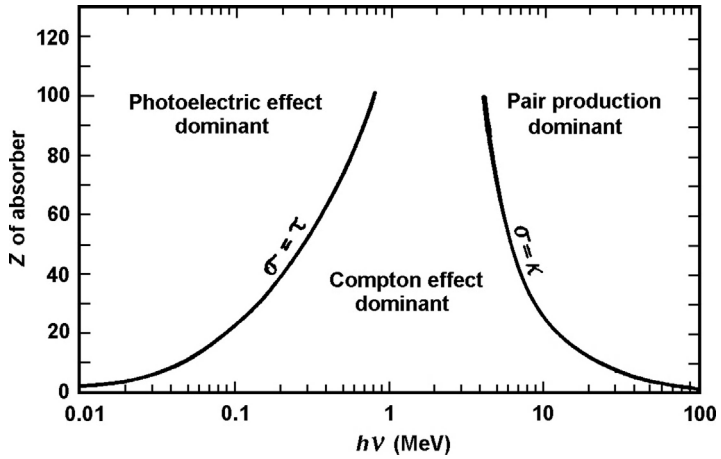


Figure 7.19 Dominance regions for the three γ-ray absorption processes, c.f. Fig. 7.17

where all energies are in MeV. The pair of electrons are absorbed as described in §7.4. The annihilation of positrons normally produce pairs of 0.51 MeV gammas, which are absorbed by the processes described previously[1].

Figure 7.19 summarizes the domains of interaction of the main γ-ray absorption processes as a function of γ-ray energy and absorber Z-value.

[1] Annihilation of positrons can in a very few cases also produce three gamma rays.

7.6. ABSORPTION OF NEUTRONS

A beam of collimated neutrons is attenuated in a thin absorber through scattering and absorption processes in a similar manner to the attenuation of γ-rays; these processes are described in previous sections. In a thick absorber the neutrons are slowed from incident energy at the absorber face to thermal energies if the absorber is thick enough. The ultimate fate of the neutron is capture in an absorber nucleus. Because of the spread in neutron energy and the energy dependency of the capture cross-sections, no simple relation can be given for the attenuation of the neutron beam (cf. next section).

7.7. RADIATION SHIELDING

The absorption properties of nuclear radiation in different materials must be known in order to design shielding to avoid unwanted radiation effects on the surroundings by nuclear radiation sources.

For charged particles the shielding is usually slightly thicker than that required for the maximum range of the particles in the material. Absorption thicknesses of 0.2 mm are adequate to completely absorb the particles from α-decay. By contrast 15 mm of materials of low Z such as water, plastic, etc., are required for absorption of β-radiation with energies up to 3 MeV. Radiation shielding constructed from materials of higher atomic number require correspondingly thinner thicknesses. The data in Table 7.2 and the curves in Figs. 7.6 and 7.13 provide information on the thickness of absorber material required for the energy of various types of radiation.

Since γ-rays and neutrons have no definite range but exhibit a logarithmic relation between thickness and intensity, only a partial reduction of the radiation can be obtained. By combining Eqns (7.7) and (7.8), Eqn (7.27) is obtained

$$\varphi = \left(n\,A/\left(4\pi\,r^2\right)\right)\mathrm{e}^{-\mu x} \tag{7.27}$$

It is seen that the intensity from a point source of radiation can be decreased by increasing either the distance from the source r or the thickness x of the absorber. Alternately, an absorber with a higher absorption coefficient, μ, can be chosen to reduce the thickness required.

Equation (7.27) is valid only for point sources with ideal geometry, i.e. no back or multiple scattering, etc. For γ-radiation the thicker the absorber the higher is the percentage of radiation which may be scattered backwards through secondary (mainly Compton) scattering. The effect of geometry and absorber thickness can be taken into account by including a constant B in the absorption equation:

$$\varphi = B\,\varphi_0\,\mathrm{e}^{-\mu x} \tag{7.28}$$

The "*dose build-up*" *factor* B not only takes into account multiple Compton and Rayleigh scattering but also includes correction for positron formation at high γ-energies